

Crusher to the Bridge: Multi-Pathogen Outbreak Simulation and Intervention Analysis Across Maritime Vessel Classes

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Technical report generated from CTB campaign outputs

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Abstract

This technical report synthesizes CTB maritime outbreak simulation campaigns spanning norovirus, respiratory pathogens, contact/fomite pathogens, environmental pathogens, and stylized Starfleet scenarios. The supplied campaign sources contain 52,150 simulation runs across C7, C8, C9, C10, C12, C12b, C12c, C14, C14b, and Trek, plus 285 CDC Vessel Sanitation Program outbreak records from 1993–2023. The main result is that the apparent VSP attack-rate flatness across ship sizes is best interpreted as a post-response observation. In C12c, norovirus dose_adj=10.6 under syndromic response produces a near-flat 5–7% attack-rate band across expedition, classic, spirit, and mega platforms. The same dose under true no-response conditions retains a physical platform gap, with expedition near 23% and mega near 11%. C12c medical response sensitivity shows that sick-call probability, detection delay, and isolation compliance have little effect once the VSP threshold backstop activates. Transport validation supports using the native analytical solver for campaign-scale attack-rate experiments: paired CONTAM/native runs showed zero agent-state deltas in the available validation summaries and in the 40 quick pairs rechecked from telemetry. C14 shows that active outbreak response saturates early. Advanced mid-voyage surveillance modalities add less than one percentage point of absolute norovirus attack-rate reduction over syndromic response, while respiratory pathogens are reduced by roughly 94–98% under basic syndromic surveillance. Trek scenarios reproduce the same principle in a stylized setting: smaller platforms with faster containment can outperform larger platforms with more advanced detection.

Keywords: agent-based model; maritime epidemiology; norovirus; SARS-CoV-2; surveillance; cruise ship; CONTAM; outbreak response

Contents

1	Introduction	5
1.1	The cruise ship as an epidemiological microcosm	5
1.2	Historical context: VSP, Diamond Princess, and MV Hondius	5
1.3	The gap addressed by this report	6
1.4	Main contributions	6
2	Platform Architecture	7
2.1	CTB simulation framework	7
2.2	Infection dynamics bridge	9
2.3	Six-pathway transmission core	9
2.4	Per-pathogen route weights as mechanistic differentiation	9
2.5	CONTAM HVAC transport versus native analytical solver	9
2.6	Ship operations layer	10
3	Vessel Platforms and Spatial Models	11
3.1	Cruise ship classes	11
3.2	Spatial layout design	11
3.3	Per-class dining parameterization	12
3.4	The expedition BuffetLido fix	13
3.5	Starship platforms	13
4	Multi-Pathogen Calibration	14
4.1	Campaign sequence	14
4.2	Ten real-world pathogens and Trek-specific pathogens	14
4.3	Dose adjustment as a shedding modulator	14

4.4	Three route-weight behavior classes	16
4.5	Density exponent experiment	16
5	The VSP Shadow	18
5.1	CDC VSP outbreak data	18
5.2	The paradox	18
5.3	Calibrating model response to VSP observations	18
5.4	The VSP threshold as the dominant intervention	20
5.5	Uncontrolled epidemic potential	22
6	Transport Solver Validation	24
6.1	CONTAM multi-zone mass balance versus native solver	24
6.2	Per-seed and per-epoch agreement	24
6.3	Environmental mass divergence	26
6.4	Why the mass difference did not alter outcomes	26
6.5	Implications for environmental surveillance	26
7	Detection and Intervention Effectiveness	27
7.1	Campaign scope	27
7.2	Surveillance stack	27
7.3	Heatmap result	29
7.4	Norovirus ceiling	29
7.5	Where advanced surveillance is likely to matter	29
7.6	Immunity interactions	29
8	Starfleet Scenarios	30
8.1	Constitution versus Galaxy class	30
8.2	Rigelian Fever and Psi 2000 Polywater	32
8.3	Barclay protomorphosis	32
8.4	Cross-era influenza comparison	32
8.5	General principle	32
9	Discussion	33
9.1	What the model reproduces	33

9.2	What the model does not yet capture	33
9.3	Boundary condition argument	33
9.4	Implications for ship design	34
9.5	Implications for public health	34
9.6	When transport fidelity matters	34
10	Future Work	35
10.1	Port visit activation and shore excursion mixing	35
10.2	Multi-voyage persistence	35
10.3	Environmental surveillance calibration	35
10.4	FOH and BOH crew segmentation	35
10.5	Embarkation health screening	36
10.6	Real-time wearable fleet surveillance	36
A	Pathogen Calibration Parameter Table	37
B	Spatial Layout Zone Inventories	38
C	Campaign Manifest Summary	39
D	VSP Outbreak Dataset	42
E	Dining Model Parameterization	43
F	Code Repository Structure and Test Coverage	44
	References and Source Notes	45

Chapter 1

Introduction

1.1 The cruise ship as an epidemiological microcosm

Cruise ships are useful controlled settings for outbreak modeling because they combine dense occupancy, repeated shared meals, stable cabin assignments, shared air-handling infrastructure, and a medical response team operating under standardized reporting rules. They are not closed systems in the strict sense. Embarkation, port calls, shore excursions, crew turnover, provisioning, and disembarkation create boundary flows. Within a voyage, however, the onboard contact graph is constrained enough to make mechanistic simulation informative.

This report treats the vessel as a spatially explicit intervention system rather than as a homogeneous population. The same pathogen can generate different attack rates on different vessel classes because dining topology, cabin density, venue modularity, and medical response scale differently with population. The central question is whether those physical differences remain visible after a standardized detection and intervention stack is activated.

1.2 Historical context: VSP, Diamond Princess, and MV Hondius

Three empirical frames motivated the work. The CDC Vessel Sanitation Program (VSP) provides outbreak surveillance for passenger vessels calling at U.S. ports. In the parsed 1993–2023 dataset used here, there are 285 outbreaks, including 211 coded as norovirus. VSP reporting is thresholded: acute gastroenteritis outbreaks become operationally visible at about 3% passenger attack rate. That threshold matters because it is not only a surveillance rule; it is also a trigger for intensified onboard response.

Diamond Princess remains the most useful high-occupancy SARS-CoV-2 calibration example in this project. The calibration target notes record 712 positives among 3,711 persons, or 19.2%, as the primary unmitigated large-vessel target. Expedition-ship respiratory outbreaks such as Greg Mortimer are higher, but they are treated here as confounded by testing and operating context

rather than as the main calibration target. The MV Hondius hantavirus scenario broadened the motivating space by emphasizing that maritime outbreak response must span pathogens that differ in route weights, incubation periods, and environmental persistence.

1.3 The gap addressed by this report

The model sequence was designed to connect four pieces that are often studied separately: a multi-platform ship model, a multi-pathogen route-weighted infection engine, high-fidelity versus native transport solvers, and a detection-to-intervention stack. The campaign set includes 52,150 simulation runs across C7, C8, C9, C10, C12, C12b, C12c, C14, C14b, and Trek, plus the 285-row VSP outbreak dataset. The primary data products are campaign-level CSV summaries and selected per-run telemetry zips.

1.4 Main contributions

The main findings are quantitative. First, norovirus dose_adj=10.6 under syndromic VSP response produces a near-flat four-platform attack-rate band: expedition 6.7%, classic 7.1%, spirit 4.6%, and mega 5.5% across all C12c syndromic sensitivity variants. Second, the corresponding uncontrolled norovirus simulations retain a physical platform gap: expedition 23.4% versus mega 10.6%, a $2.2\times$ ratio. Third, sick-call probability, detection delay, and isolation compliance have small effects at dose 10.6 once the VSP threshold backstop exists. Fourth, the CONTAM and native transport solvers generate identical agent outcomes in paired validation runs, even when environmental mass differs. Fifth, C14 indicates that mid-voyage advanced surveillance has little marginal value after syndromic detection for active outbreaks, while respiratory pathogens are reduced by 93.8–97.8% under syndromic surveillance alone.

Chapter 2

Platform Architecture

2.1 CTB simulation framework

Crusher to the Bridge (CTB) is an epoch-based agent simulation. Each run initializes a vessel platform, a population, a pathogen configuration, a surveillance strategy, and run-level random seed. The engine then advances hourly epochs over a 168-hour voyage. During each epoch, agents occupy zones, interact through contact and environmental pathways, accumulate pathogen dose, change infection state, and may enter medical response workflows.

The simulation loop is organized around reproducibility. Campaign CSVs store run identifiers, pathogen, platform, population, dose adjustment, density exponent, immunity, surveillance mode, seed, attack rate, peak prevalence, outbreak occurrence, and related metrics when available. Per-run zip archives store run specifications, summaries, and compact timeseries telemetry.

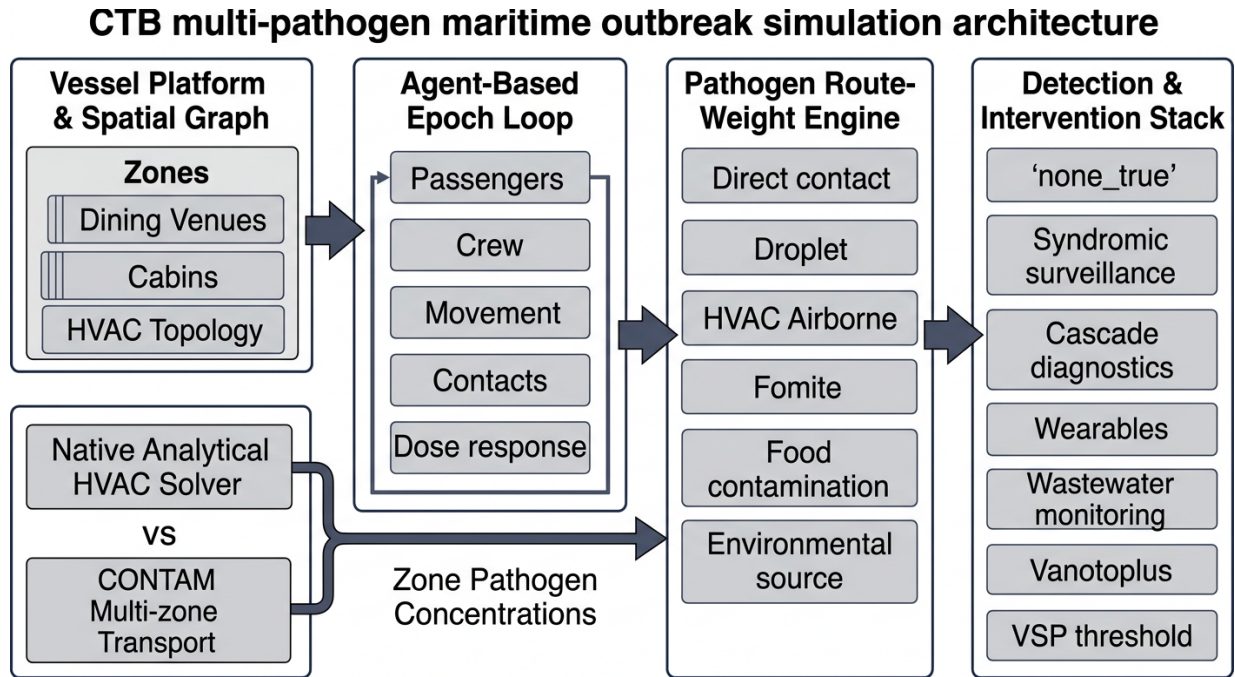


Figure 2.1: System architecture of the CTB maritime outbreak simulation. The figure shows the vessel platform layer, the epoch-level agent loop, the route-weighted pathogen engine, the detection/intervention stack, and the native-versus-CONTAM transport branch.

2.2 Infection dynamics bridge

The infection dynamics bridge inherited the Korkin Lab-style agent-state structure but was extended to support per-pathogen route weights. Dose adjustment, written as `dose_adj`, is used as a shedding and infectious-dose modulator during calibration. Lower `dose_adj` values generally produce higher attack rates for the dose-responsive pathogens in these campaigns. The effect is pathogen-specific because route weights, immunity, and intervention timing differ.

2.3 Six-pathway transmission core

The transmission core includes direct contact, droplet, HVAC airborne, fomite, food contamination, and environmental-source pathways. These are not six independent diseases; they are weighted route contributions for a single pathogen profile. Norovirus in the calibrated configuration emphasizes direct contact, fomite, and food contamination. Respiratory pathogens such as influenza, measles, and SARS-CoV-2 emphasize droplet and HVAC airborne routes. Environmental and food-borne profiles such as *Legionella* and *Vibrio* behave differently in the dose sweeps because the modeled exposure source is less sensitive to person-to-person amplification.

2.4 Per-pathogen route weights as mechanistic differentiation

C8 used ten pathogens with route weights and showed that route composition produces distinct response classes. Airborne-dominant pathogens had little platform gap in the early three-platform sweeps. Contact/fomite pathogens retained vessel-structure sensitivity. Environmental or food-borne pathogens were often dose-insensitive over the tested range. This was the rationale for carrying route-weighted pathogens into C12 and C14 rather than calibrating all pathogens with a single contact kernel.

2.5 CONTAM HVAC transport versus native analytical solver

The HVAC layer can be evaluated through a native analytical star-topology solver or a CONTAM-derived multi-zone transport path. The native solver is faster and is therefore the practical choice for campaign-scale experiments. The validation question is whether solver differences change agent outcomes. Chapter 6 reports paired validation results: agent outcomes were identical in the tested pairs, while environmental pathogen mass differed by about 1.3–1.7 $\times$ in median comparisons depending on pathogen.

2.6 Ship operations layer

The ship operations layer adds voyage itinerary, port-visit hooks, and per-class meal weights. Port effects are currently represented as configurable stubs that default to an all-sea-day voyage, so the campaigns reported here isolate onboard dynamics. The per-class meal weights are active and materially changed the calibration problem because dining choice changes the contact graph for all pathogens, not only norovirus.

Chapter 3

Vessel Platforms and Spatial Models

3.1 Cruise ship classes

The four cruise platforms represent increasing vessel size and venue complexity: expedition (450 agents), classic (1,910 agents), spirit (3,000 agents), and mega (7,000 agents). The exact class names are model classes rather than ship brands. They encode occupancy, zone inventories, dining architecture, medical response gradients, and HVAC/spatial topology.

3.2 Spatial layout design

Each platform is a zone graph. Cabins, dining venues, public spaces, crew areas, medical spaces, and HVAC-connected zones form the spatial substrate. The dining graph receives special attention because norovirus and other gastrointestinal pathogens are sensitive to shared surfaces, fixed versus open seating, and food-service modality. Table 3.1 summarizes the platform assumptions, and Figure 3.1 shows the class-level dining layouts used after the ship-operations update.

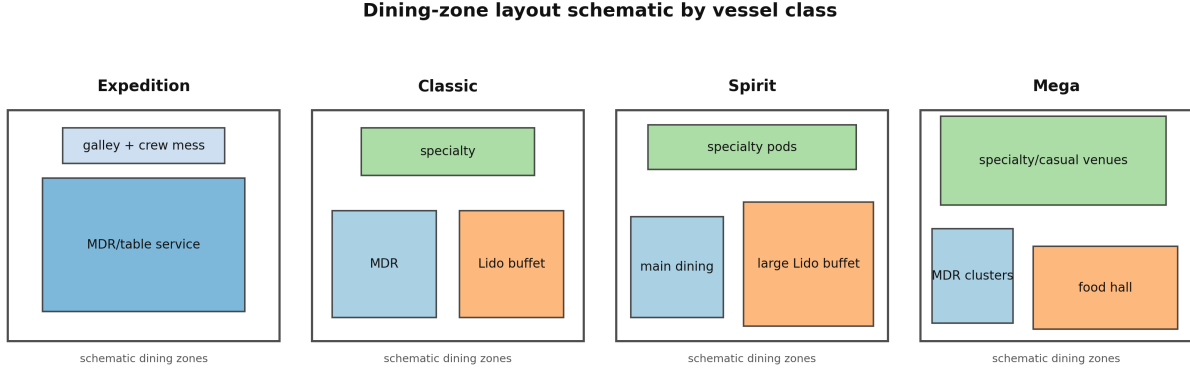


Figure 3.1: Dining-zone schematics. Expedition uses crew-served table service with no self-service buffet. Classic, spirit, and mega platforms include buffet exposure gradients and increasing specialty/casual venue decentralization.

Platform	Agents	Dining model	Medical response defaults
Expedition	450	Table service; no self-service buffet; breakfast/lunch routed to MDR and crew-served venues.	sick call 0.90; compliance 0.95; delay 1 h
Classic	1,910	Premium-style MDR, Lido buffet, and specialty dining; buffet exposure highest at breakfast and lunch.	sick call 0.70; compliance 0.90; delay 2 h
Spirit	3,000	Contemporary buffet gradient with high breakfast/lunch buffet allocation and MDR-dominant dinner.	sick call 0.50; compliance 0.85; delay 2 h
Mega	7,000	Decentralized food hall plus specialty/casual venues; lower per-venue cohort overlap.	sick call 0.40; compliance 0.80; delay 3 h

Table 3.1: Platform comparison used in C12 onward. Dining weights are listed in text; response defaults come from the per-platform medical-response specification.

3.3 Per-class dining parameterization

The dining parameterization maps the report-derived vessel classes to model classes. Expedition maps to the small-ship table-service case. Classic maps to premium vessels. Spirit maps to contemporary vessels. Mega maps to mega/iconic vessels. The meal weights are: expedition breakfast MDR 0.85 and crew mess 0.15, lunch MDR 0.70 and crew mess 0.30, dinner MDR 0.90, specialty 0.05, and crew mess 0.05; classic breakfast buffet 0.75, MDR 0.20, specialty 0.05, lunch buffet 0.60, MDR 0.25, specialty 0.15, dinner buffet 0.20, MDR 0.60, specialty 0.20; spirit breakfast buffet 0.80,

MDR 0.18, specialty 0.02, lunch buffet 0.65, MDR 0.25, specialty 0.10, dinner buffet 0.25, MDR 0.65, specialty 0.10; mega breakfast buffet 0.70, MDR 0.15, specialty 0.15, lunch buffet 0.50, MDR 0.20, specialty 0.30, dinner buffet 0.20, MDR 0.45, specialty 0.35.

3.4 The expedition BuffetLido fix

The expedition fix was mechanistic. Earlier layouts routed expedition casual dining through a buffet-like venue. The operations review showed that expedition food service is table-service dominated, with high food-and-beverage crew ratios and minimal unmonitored self-service. C10 verified this fix on the expedition platform with 1,140 runs. The fix did not force the expedition class to match larger ships; rather, it corrected the exposure mechanism so that small-vessel norovirus behavior emerged from contact topology rather than from an inappropriate self-service buffet assumption.

3.5 Starship platforms

The Trek campaign uses the same physical and intervention logic in two stylized platforms: Constitution class with 430 crew and Galaxy class with 1,014 crew plus families. These platforms are not empirical cruise ships. They are stress tests for whether platform size, contact topology, and medical response timing matter even when detection technology changes.

Chapter 4

Multi-Pathogen Calibration

4.1 Campaign sequence

C7 recalibrated norovirus and SARS-CoV-2 across three platforms in 5,250 runs. C8 added ten real-world pathogens with route weights in 2,295 runs. C9 performed fine calibration and a density-exponent sweep in 8,745 runs. C10 verified the expedition dining correction in 1,140 runs. C12 then re-established calibration after the ship-operations layer in 6,720 runs.

4.2 Ten real-world pathogens and Trek-specific pathogens

The real-world pathogen set in C8 included norovirus, SARS-CoV-2, influenza, measles, *C. difficile*, *Campylobacter*, *Legionella*, *Vibrio*, hantavirus, and Ebola. The Trek campaign added four scenario pathogens: Rigelian fever, Psi 2000 polywater, Barclay protomorphosis, and influenza in a cross-era configuration.

4.3 Dose adjustment as a shedding modulator

The dose-response panels in Figure 4.1 show that `dose_adj` is not a universal attack-rate dial. *Campylobacter*, *C. difficile*, hantavirus, influenza, measles, SARS-CoV-2, and norovirus were dose responsive over at least part of their tested ranges. *Legionella* remained near 2.1% across its tested doses. *Vibrio* remained near 10.7%. Ebola saturated at 100% in the C8 range.

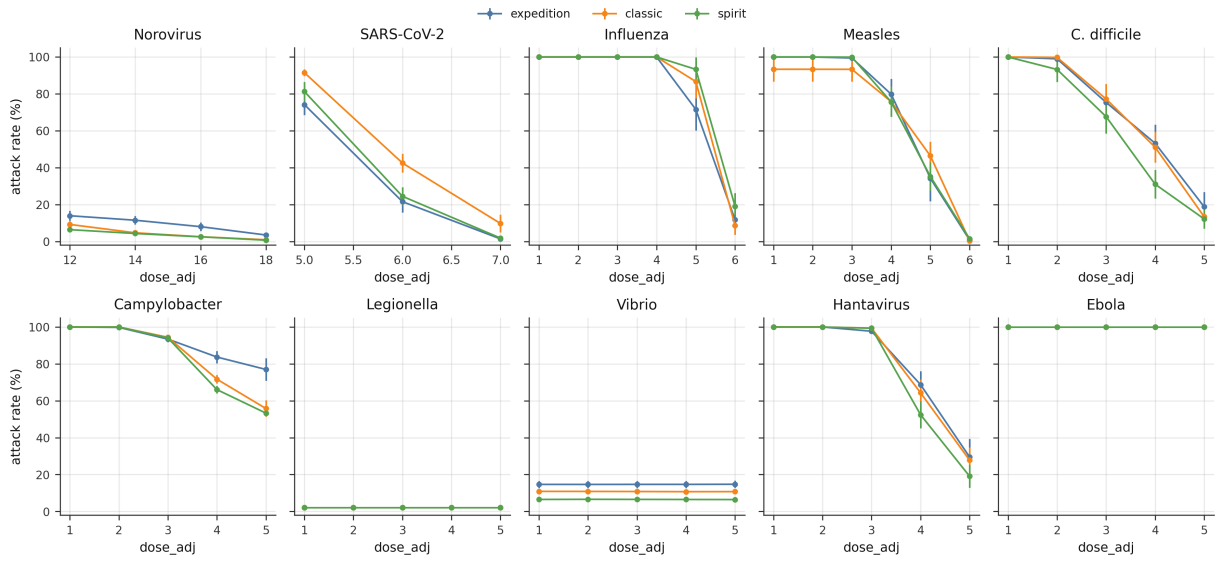


Figure 4.1: C8 ten-pathogen dose-response panel across expedition, classic, and spirit platforms. Points show mean attack rate by dose_adj and platform; error bars show standard error across seeds.

4.4 Three route-weight behavior classes

The C8 and C12 results support three practical behavior classes. Airborne-dominant pathogens, represented by SARS-CoV-2, influenza, and measles, are strongly controlled by early detection and isolation once symptomatic signal is available. Contact/fomite pathogens, represented by norovirus and *C. difficile*, are more sensitive to dining and surface topology. Environmental or foodborne profiles, represented in the C8 range by *Legionella* and *Vibrio*, are less responsive to person-to-person contact scaling and need source-process calibration rather than only dose_adj tuning.

4.5 Density exponent experiment

C9 tested whether changing the contact-density exponent α could remove the gastrointestinal platform gap. At norovirus dose 14 and immunity 0, expedition attack rate stayed between 10.4% and 10.6% across $\alpha = 0.50$ to 1.00, while spirit stayed between 3.8% and 3.9%. The expedition/spirit ratio stayed near $2.7\times$. Thus, the gap was not an artifact of one density exponent.

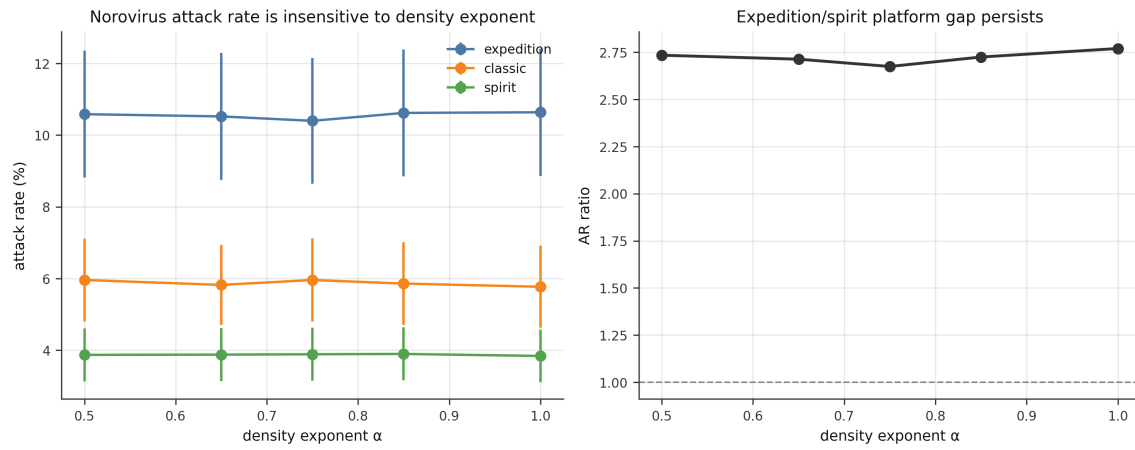


Figure 4.2: C9 alpha sweep for norovirus dose 14. The platform gap persists across the tested density exponents, indicating that dining and spatial topology, not only density scaling, generate the observed small-vessel excess.

Chapter 5

The VSP Shadow

5.1 CDC VSP outbreak data

The parsed VSP dataset contains 285 outbreaks from 1993–2023. Norovirus accounts for 216 rows. Passenger totals range from 18 to 6285, and passenger attack rates average 6.2% across all outbreak causes. For norovirus-coded outbreaks above 500 passengers, median attack rates stay close to 5–6% across size bins (Table C.4).

5.2 The paradox

The VSP attack-rate pattern appears flat across ship sizes, even though background acute gastroenteritis incidence in external notes was higher on larger ships. The initial interpretation was that the model’s small-vessel excess was wrong. The corrected interpretation is that VSP outbreak data are post-intervention observations. Once a ship reaches the VSP response threshold, cleaning, case isolation, reporting, and service changes compress the final attack-rate distribution.

5.3 Calibrating model response to VSP observations

C12c provides the clearest calibration point. At norovirus dose_adj=10.6 under all syndromic variants, mean attack rates were expedition 6.7%, classic 7.1%, spirit 4.6%, and mega 5.5%. Under the stock syndromic configuration alone, the same values were expedition 6.3%, classic 7.0%, spirit 5.1%, and mega 5.1%. Figure 5.1 shows the fine sweep from dose 10.0 to 11.0.

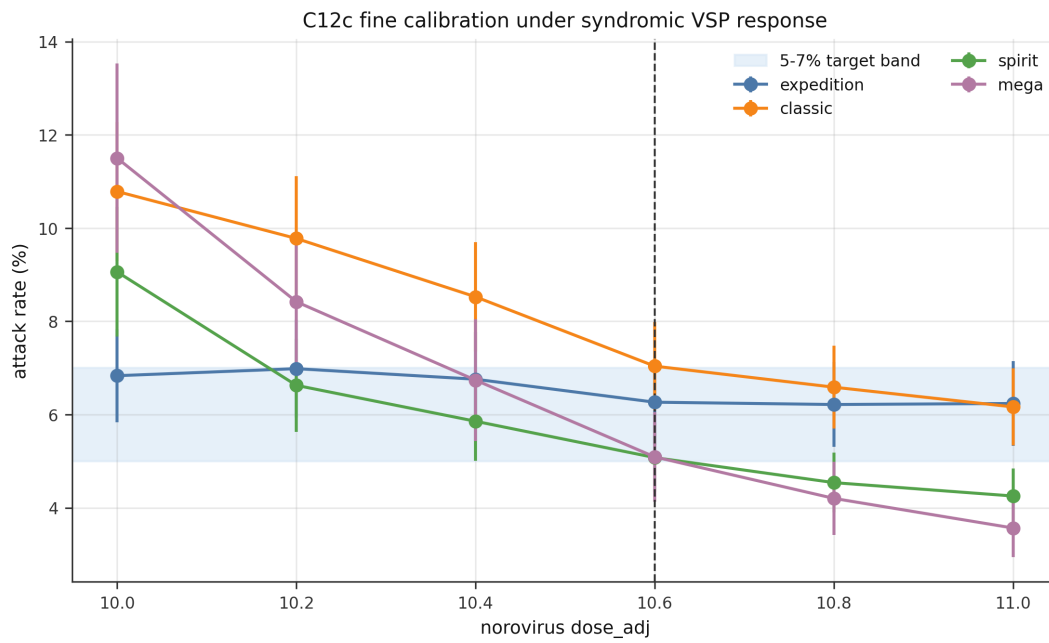


Figure 5.1: C12c fine calibration of norovirus under syndromic response. Dose_adj=10.6 places the four platforms near the 5–7% VSP-consistent band, with spirit slightly below the band when averaged across all syndromic sensitivity variants.

5.4 The VSP threshold as the dominant intervention

The medical response sensitivity analysis varied sick-call probability, detection delay, and isolation compliance. At dose 10.6, the minimum-to-maximum ranges across these variants were narrow relative to the uncontrolled gap: expedition 6.6–6.9%, classic 6.9–7.5%, spirit 4.3–4.7%, and mega 5.1–6.4%. The VSP backstop dominates because the threshold response activates once cumulative symptomatic burden crosses the operational trigger.

C14 independently supports the same mechanism. Norovirus mean attack rate was 15.9% under true no-response conditions, 10.1% with syndromic detection but threshold off, 7.1% at a 5% threshold, 5.1% at the 3% threshold, and 5.0% at the 1% threshold.

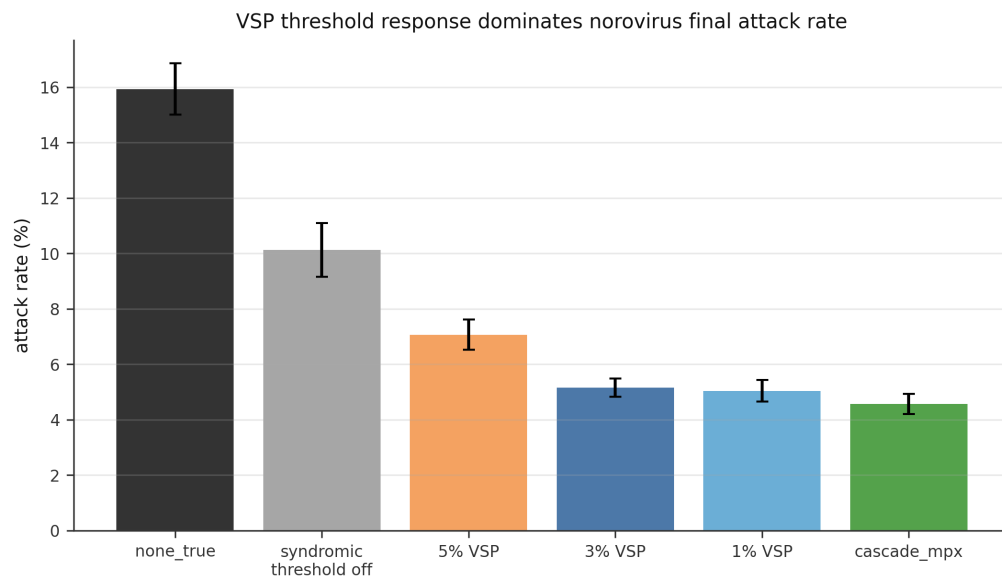


Figure 5.2: C14 VSP threshold sweep. Lower threshold settings compress final norovirus attack rates. The 3% threshold recovers the VSP-consistent range without requiring platform-specific tuning.

5.5 Uncontrolled epidemic potential

The true no-response experiment can only be run in silico. At `dose_adj=10.6` in C12c, uncontrolled attack rates were expedition 23.4%, classic 22.6%, spirit 12.2%, and mega 10.6%. The expedition/mega ratio was $2.2\times$ using C12c means; the same statement is approximately the $2.3\times$ platform gap used throughout the campaign notes. Figure 5.3 overlays VSP outbreak points with syndromic and no-response model means.

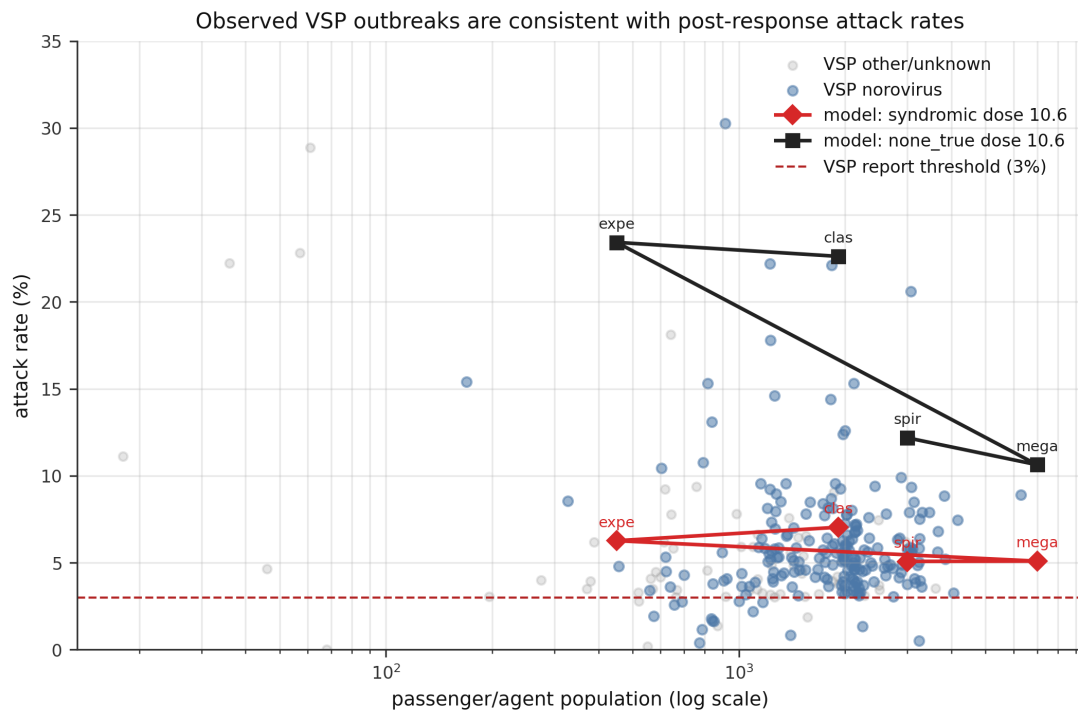


Figure 5.3: VSP outbreak attack rates and model overlay. VSP points are observed post-response outbreaks. The syndromic dose 10.6 model line lies near the observed norovirus band, whereas the true no-response line retains the physical platform gap.

Chapter 6

Transport Solver Validation

6.1 CONTAM multi-zone mass balance versus native solver

The validation compared CONTAM-derived multi-zone transport with the native analytical solver. The native solver is intended for campaign-scale simulation. CONTAM provides the higher-fidelity reference path for HVAC transport. The practical question was whether transport-solver differences propagate into agent infection outcomes.

6.2 Per-seed and per-epoch agreement

The paired validation summary available in project memory reports 200/200 paired runs with zero deltas across agent outcome metrics. The 200 pairs covered norovirus, influenza, measles, and SARS-CoV-2. The reported metric set included attack rate, peak prevalence, outbreak occurrence, peak epoch, detection epoch, detection lag, effective reproduction proxy, quarantine-person-epochs, and final susceptible fraction. For the norovirus quick pairs regenerated here from the available telemetry, 40 matching CONTAM/native zip pairs produced zero agent-state deltas across susceptible, infected, symptomatic, recovered, immune, quarantined, isolated, and new infection timeseries fields.

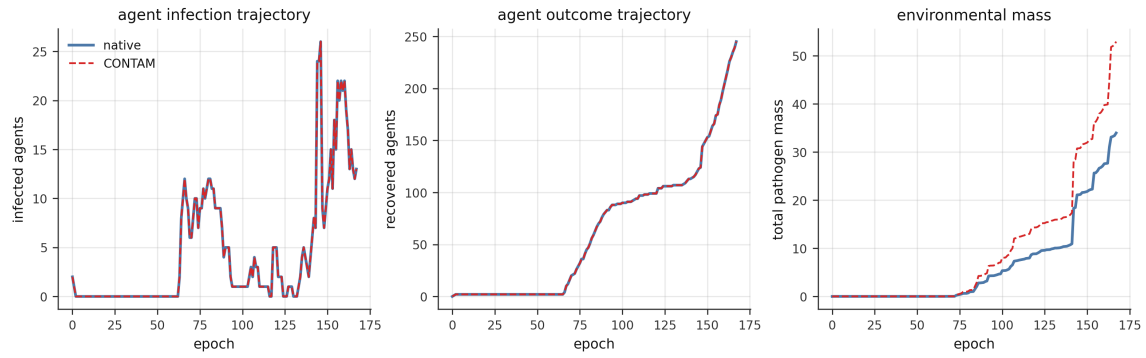


Figure 6.1: CONTAM versus native timeseries overlay for a representative norovirus dose 10.6 run. Agent-state trajectories are identical for infected and recovered counts. Environmental pathogen mass differs, consistent with the broader validation summary, but did not change agent outcomes in the tested pairs.

6.3 Environmental mass divergence

The broader validation notes report pathogen-mass median ratios of about $1.67\times$ for norovirus, $1.42\times$ for SARS-CoV-2, $1.41\times$ for influenza, and $1.28\times$ for measles. The 40 quick pairs extracted for this report had median mass ratio $1.57\times$. Some per-epoch ratios can be very high when the native denominator is near zero, so median ratios are more informative than maxima.

6.4 Why the mass difference did not alter outcomes

There are three likely explanations. First, dose-response saturation can make a two-fold zone-mass difference irrelevant once the susceptible agent’s dose is already on the high-probability part of the curve. Second, route-weight scaling occurs after zone-level calculations, so environmental mass may not linearly map to infection probability. Third, total shipboard mass can differ in zones that contribute little to occupied-agent dose. The validation result supports using the native solver for intervention campaigns, while reserving CONTAM for calibration of environmental surveillance and airflow-specific engineering questions.

6.5 Implications for environmental surveillance

The equivalence result should not be overextended. Agent infection outcomes were identical in the tested outbreak simulations, but environmental surveillance depends on measured mass, concentration, and sampling location. Wastewater, surface swab, and air-sampling calibration may therefore require solver-specific correction even when outbreak attack rates do not.

Chapter 7

Detection and Intervention Effectiveness

7.1 Campaign scope

The intervention evidence comes primarily from C14 (8,440 runs) and C14b (4,220 runs), with C12c providing the norovirus threshold and medical-response sensitivity anchor. Across the full project set summarized in this report, the campaign table contains 52,150 runs. The outline target of 52,390 total runs is therefore higher than the CSV and zip sources supplied for this monograph; this report uses the supplied sources and flags the discrepancy rather than filling the gap.

7.2 Surveillance stack

The modeled stack runs from no response to increasingly instrumented detection: `none_true`, `syndromic`, `cascade`, `cascade_mpx`, wearable variants, and wastewater. C14 also includes VSP threshold variants for norovirus. The stack is evaluated during active onboard outbreaks, not at embarkation screening.

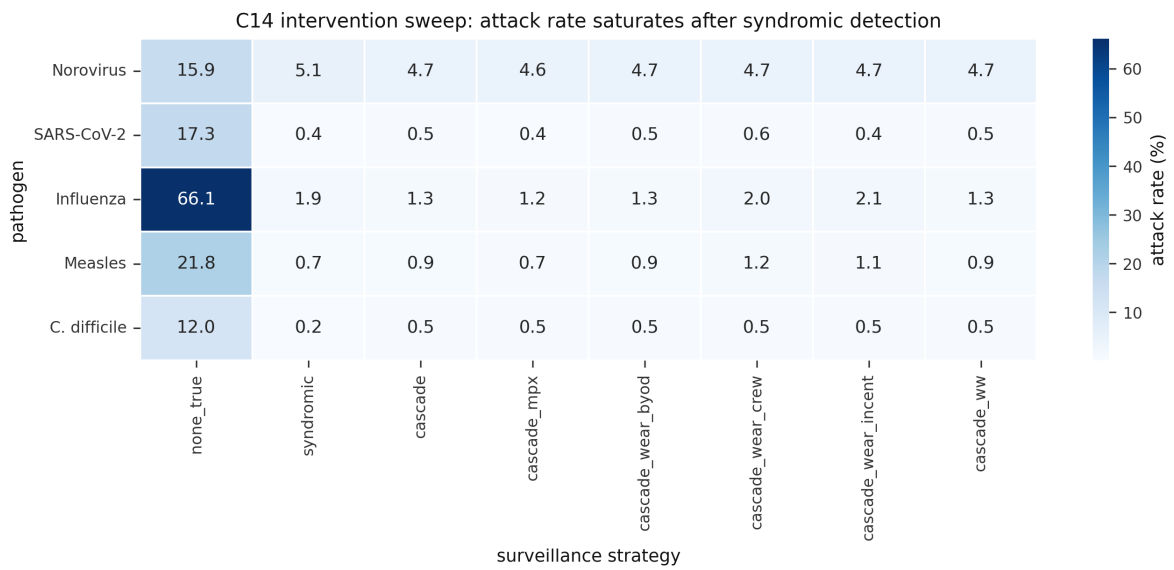


Figure 7.1: C14 surveillance heatmap. Values are mean attack rates by pathogen and strategy. For active outbreaks, most benefit occurs by the syndromic stage; advanced modalities add little for norovirus and small marginal changes for the respiratory pathogens in this configuration.

7.3 Heatmap result

Figure 7.1 shows that response saturates early. Norovirus fell from 15.5% baseline to 4.9% under syndromic surveillance, a 66.6% reduction. Cascade multiplex reduced it to 4.6%. The incremental gain over syndromic was less than 1 percentage point in absolute attack rate.

For respiratory pathogens, any active surveillance strategy was highly effective. Under syndromic surveillance alone, SARS-CoV-2 was reduced by 97.8%, influenza by 97.3%, and measles by 93.8%. The broader range across respiratory pathogens and surveillance strategies was 91.5–98.1%.

7.4 Norovirus ceiling

Norovirus remained the least controllable pathogen in the C14 set. Its C14 syndromic reduction was about two-thirds, and the best listed mean among the major active strategies was cascade multiplex at 4.6%. The outline described a 62% reduction ceiling; the supplied C14 CSV gives a stronger mean reduction of 69.6% for cascade multiplex and 66.6% for syndromic. The text reports the CSV-derived values.

7.5 Where advanced surveillance is likely to matter

The mid-voyage active-outbreak experiments suggest limited incremental value for wearables and wastewater once symptomatic response and threshold backstop operate. That does not imply low value at the boundary. Embarkation, turnaround, crew carryover, and port-return screening are plausible use cases because they act before onboard amplification. Those boundary interventions were not tested in the supplied C14 CSV.

7.6 Immunity interactions

Immunity changes the baseline and therefore changes how much surveillance can improve outcomes. In C14, measles includes an 85% immunity condition and SARS-CoV-2 includes a 30% immunity condition. When immunity already suppresses baseline transmission, surveillance has less remaining attack rate to remove. This is a boundary condition on intervention interpretation, not a contradiction of the main active-response result.

Chapter 8

Starfleet Scenarios

8.1 Constitution versus Galaxy class

The Trek campaign applies the same simulation concepts to two stylized platforms: Constitution class and Galaxy class. The Constitution platform has 430 agents. The Galaxy platform has 1,014 agents and a larger, family-containing contact topology. The comparison is useful because it separates detection technology from vessel topology in a deliberately artificial setting.

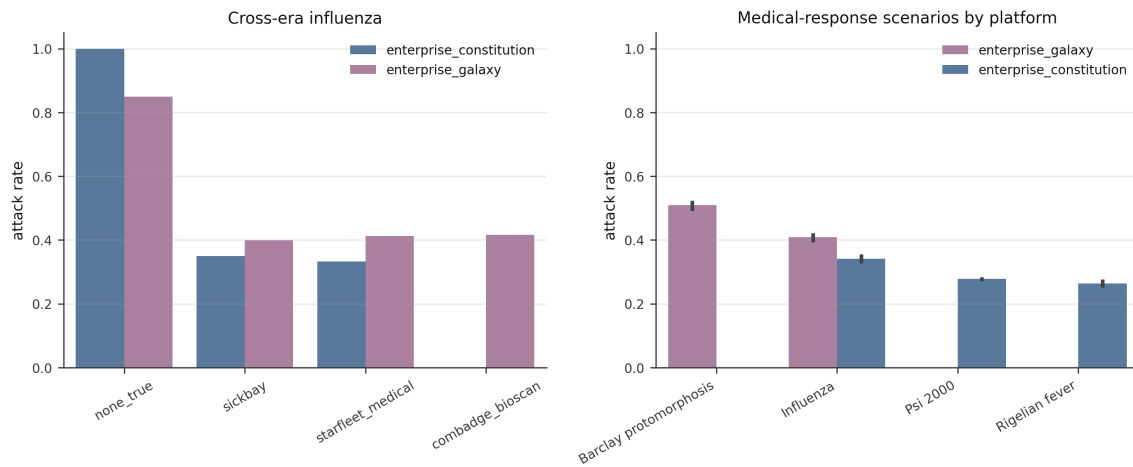


Figure 8.1: Trek campaign cross-era comparison. Constitution-class influenza under sickbay or Starfleet medical response has lower mean attack rate than Galaxy-class influenza under the corresponding response classes. Barclay protomorphosis remains high even with advanced bioscan-like detection.

8.2 Rigelian Fever and Psi 2000 Polywater

Rigelian fever and Psi 2000 polywater were simulated on the Constitution platform. In the supplied CSV, both reach 100% mean attack rate under `none_true`. With sickbay or Starfleet medical response, Rigelian fever falls to about 25.7–27.1% and Psi 2000 to about 27.7–27.8%. These are large reductions, but they do not eliminate the outbreak.

8.3 Barclay protomorphosis

Barclay protomorphosis was simulated on the Galaxy platform. It reaches 100% under `none_true`. With sickbay, Starfleet medical, or combadge bioscan strategies, mean attack rates remain about 49.7–51.7%. This supports the outline claim that even combadge-style bioscan cannot stop this scenario in the tested configuration.

8.4 Cross-era influenza comparison

The influenza comparison tests the general platform-size argument. Constitution-class influenza has 100% mean attack rate under `none_true` and falls to 34.9% under sickbay and 33.3% under Starfleet medical response. Galaxy-class influenza has 85.0% under `none_true` and remains at 39.9–41.6% under sickbay, Starfleet medical, or combadge bioscan. The smaller ship with faster containment outperforms the larger ship with advanced technology in this campaign.

8.5 General principle

The Trek scenarios should not be read as empirical claims about fictional medicine. They are sensitivity experiments. The result is consistent with the cruise-ship campaigns: platform size and contact topology can dominate detection modality once the response system has already reached an effective threshold.

Chapter 9

Discussion

9.1 What the model reproduces

The model reproduces an important empirical pattern after the calibration frame is corrected. VSP observations are post-response data. Calibrating no-response simulations to those observations conflates intrinsic epidemic potential with intervention-compressed outcomes. When `model(syndromic)` is compared to VSP, norovirus dose 10.6 produces a flat 5–7% attack-rate band across the four platforms. When `model(none_true)` is run at the same dose, the physical platform gap returns.

9.2 What the model does not yet capture

Several mechanisms remain represented as stubs or simplified assumptions. Port visits are configurable but inactive in the reported campaigns. Shore-excursion mixing, reembarkation exposure, and port-specific infection pressure are not active. Multi-voyage persistence is not modeled as a coupled environmental reservoir plus crew carryover problem. Crew social networks are simplified. Passenger behavior is less heterogeneous than real ships, especially in risk tolerance, sick-call behavior, and willingness to comply with isolation. Dining assignment is parameterized by class-level meal weights rather than by empirical individual reservation histories.

9.3 Boundary condition argument

The active-outbreak intervention results point to a boundary argument. Once onboard spread has produced symptomatic signal and activated threshold response, advanced mid-voyage surveillance adds little in the tested configuration. The same technologies may be more valuable before amplification: at embarkation, during turnaround, when screening crew carryover risk, or when port-return groups re-enter the ship. This distinction should drive future campaign design.

9.4 Implications for ship design

Dining venue architecture is an infection-control parameter. The expedition BuffetLido correction changed the interpretation of small-vessel norovirus behavior because self-service exposure was no longer incorrectly assigned. Larger ships can dilute contacts through modular venues, but they also add more shared food-service interfaces. The balance depends on fixed versus open dining, buffet self-service, staff-mediated service, and the degree of cohort mixing.

9.5 Implications for public health

The VSP threshold response appears highly cost-effective in simulation because it compresses final attack rates without needing high-end detection modalities. A lower response threshold reduces mean attack rate further, but lower thresholds may impose more isolation, service disruption, and operational cost. The present report quantifies attack-rate effects, not passenger welfare or economic cost.

9.6 When transport fidelity matters

The CONTAM/native equivalence result is strong for agent outcomes in the tested runs. It is weaker for environmental surveillance and engineering design. If the endpoint is attack rate under saturated dose-response conditions, the native solver is adequate for campaign sweeps. If the endpoint is sampled concentration in a specific zone, HVAC engineering changes, or sensor calibration, the environmental mass differences must be handled directly.

Chapter 10

Future Work

10.1 Port visit activation and shore excursion mixing

The itinerary layer includes port-day configuration fields, but the reported campaigns default to all-sea-day behavior. Future campaigns should activate disembarkation fractions, shore infection probability, reembarkation windows, and port-day dining multipliers. This will test whether port days interrupt onboard transmission, add exogenous infections, or both.

10.2 Multi-voyage persistence

Cruise outbreaks can span more than one voyage through contaminated environments or infected crew. A multi-voyage model should carry environmental reservoirs, asymptomatic crew infections, galley contamination state, and turnaround cleaning effectiveness across sailings.

10.3 Environmental surveillance calibration

The solver-validation result creates a specific follow-up. Environmental mass differs between transport solvers even when agent outcomes match. Wastewater, air sampling, and surface swab models need concentration calibration using telemetry, not only attack-rate validation.

10.4 FOH and BOH crew segmentation

Food-service transmission depends on front-of-house and back-of-house roles. Crew segmentation should separate galley preparation, buffet service, MDR waitstaff, cabin stewards, medical staff, and general hotel operations. That segmentation would support targeted screening and galley-mediated outbreak scenarios.

10.5 Embarkation health screening

Advanced surveillance may be more valuable at the boundary than mid-voyage. Embarkation screening campaigns should test symptom checks, rapid tests, wastewater from terminal facilities, wearable preboarding data, and risk-adapted boarding holds.

10.6 Real-time wearable fleet surveillance

Wearables may support fleet-level decisions even if they add little during an active onboard outbreak. Future campaigns should evaluate voyage termination, itinerary modification, targeted cleaning, and cross-voyage risk scoring rather than only final attack rate.

Appendix A

Pathogen Calibration Parameter Table

The supplied campaign CSVs do not include the full route-weight table for all fourteen pathogens as a single structured file. The route-weight concepts are described in the main text, and the dose-adjustment ranges are visible in the campaign manifest and C8/C12 runs. A complete appendix should be generated from the pathogen configuration files in the code repository when this report is synchronized with the model source tree.

Appendix B

Spatial Layout Zone Inventories

The supplied source set includes platform descriptions and dining parameterization but not a complete zone-inventory export for all six platforms. Figure [3.1](#) gives the class-level dining layouts used for interpretation. A future build should include a machine-exported zone count table by platform and zone type.

Appendix C

Campaign Manifest Summary

Table C.1: Campaign manifest summary for C7–C14, C14b, and Trek.

Campaign	Runs	Purpose	Pathogens	Platforms
C7	5250	3-platform norovirus + SARS-CoV-2 recalibration	norovirus, SARS-CoV-2	expedition, classic, spirit
C8	2295	10 pathogens with route weights	10 real-world pathogens	expedition, classic, spirit
C9	8745	fine calibration + alpha sweep	7 dose-responsive pathogens	expedition, classic, spirit
C10	1140	expedition buffet fix verification	norovirus, SARS-CoV-2	expedition
C12	6720	full recalibration with per-class meal weights	7 dose-responsive pathogens	4 cruise classes
C12b	3900	syndromic calibration sweep	norovirus, SARS-CoV-2	4 cruise classes
C12c	5920	fine calibration + medical response sensitivity	norovirus	4 cruise classes
C14	8440	advanced surveillance sweep	norovirus, SARS-CoV-2, influenza, measles, C. difficile	4 cruise classes
C14b	4220	doubled small-vessel surveillance coverage	5 pathogens	expedition, classic
Trek	5520	Starfleet scenario sweep	4 scenario pathogens	Constitution, Galaxy

Table C.2: Norovirus dose 10.6 calibration and medical-response sensitivity in C12c.

Platform	none_true(%)	synd mean (%)	synd_stock (%)	med min	med max
expedition	23.42	6.65	6.26	6.55	6.94
classic	22.62	7.10	7.04	6.90	7.49

Platform	none_true(%)	synd mean (%)	synd_stock (%)	med min	med max
spirit	12.18	4.60	5.08	4.27	4.73
mega	10.63	5.52	5.09	5.08	6.40

Table C.3: Matched C14 intervention reductions by pathogen and selected strategy.

Pathogen	Strategy	baseline (%)	post AR (%)	reduction (%)
C. difficile	syndromic	12.02	0.18	98.54
influenza	syndromic	66.08	1.89	97.26
measles	syndromic	21.78	0.75	93.84
norovirus	syndromic	15.51	4.91	66.65
SARS-CoV-2	syndromic	17.29	0.38	97.81
C. difficile	cascade_mpx	12.02	0.54	93.80
influenza	cascade_mpx	66.08	1.20	98.09
measles	cascade_mpx	21.78	0.67	93.47
norovirus	cascade_mpx	15.51	4.56	69.65
SARS-CoV-2	cascade_mpx	17.29	0.37	97.88
C. difficile	cascade_ww	12.02	0.53	93.93
influenza	cascade_ww	66.08	1.28	97.91
measles	cascade_ww	21.78	0.89	92.34
norovirus	cascade_ww	15.51	4.73	68.22
SARS-CoV-2	cascade_ww	17.29	0.54	96.92

Table C.4: CDC VSP norovirus outbreak attack rates by passenger count bin.

passenger bin	mean AR (%)	median AR (%)	outbreaks
<500	17.94	11.98	4
500–1500	6.15	5.09	70
1500–3000	5.91	5.22	111
3000–5000	6.50	5.78	30
>5000	8.91	8.91	1

Table C.5: Trek scenario attack rates by pathogen, platform, and medical strategy.

Pathogen	Platform	Strategy	AR (%)	count
Barclay protomorphosis	Galaxy	combadge_bioscan	51.75	240
Barclay protomorphosis	Galaxy	none_true	100.00	240
Barclay protomorphosis	Galaxy	sickbay	49.68	240
Barclay protomorphosis	Galaxy	starfleet_medical	51.34	240
influenza	Constitution	none_true	100.00	180

Pathogen	Platform	Strategy	AR (%)	count
influenza	Constitution	sickbay	34.91	180
influenza	Constitution	starfleet_medical	33.27	180
influenza	Galaxy	combadge_bioscan	41.57	600
influenza	Galaxy	none_true	85.00	600
influenza	Galaxy	sickbay	39.95	600
influenza	Galaxy	starfleet_medical	41.26	600
Psi 2000 polywater	Constitution	none_true	100.00	240
Psi 2000 polywater	Constitution	sickbay	27.72	240
Psi 2000 polywater	Constitution	starfleet_medical	27.82	240
Rigelian fever	Constitution	none_true	100.00	300
Rigelian fever	Constitution	sickbay	25.69	300
Rigelian fever	Constitution	starfleet_medical	27.15	300

Appendix D

VSP Outbreak Dataset

The VSP outbreak dataset contains 285 rows with ship, pathogen, passenger total, passenger attack rate, and passenger cases. The binned norovirus summary appears in Table [C.4](#). The full row-level dataset is not reproduced in the monograph body to avoid a long table that would obscure the main calibration result.

Appendix E

Dining Model Parameterization

The dining model is summarized in Chapter 3. The operational source report describes expedition, premium, contemporary, and mega/iconic food-service structures, including gross tonnage ranges, passenger capacity ranges, food-and-beverage crew ratios, venue counts, buffet capacity share, MDR share, and specialty share. These values were mapped into the model meal weights listed in Chapter 3 and used in C12 onward.

Appendix F

Code Repository Structure and Test Coverage

The repository contains CONTAM interoperability documentation and transport comparison tests, including dual-engine compare tests and platform-specific CONTAM hobbyist tests. The monograph sources did not include a formal test-coverage export, so no numeric coverage percentage is reported.

References and Source Notes

This preprint-style technical report is grounded in the supplied CTB campaign CSVs and context documents. Campaign IDs in the text refer to C7, C8, C9, C10, C12, C12b, C12c, C14, C14b, and Trek. The empirical outbreak comparison uses the parsed CDC VSP outbreak table supplied with the project. External literature metadata were not supplied as BibTeX with this artifact-generation task, so the report does not add unsupported bibliographic citations.